Lecture 26 — Injective and Surjective Functions

CS 173: Discrete Structures · Functions

University of Illinois at Urbana-Champaign

1 Setup

Throughout this lecture, let A and B be sets and let $f : A \to B$ be a function.

2 Injective Functions (Injections)

Definition 2.1 (Injective function / injection)**.** f is *injective* (also called a *one-to-one function* or an *injection*) if and only if

$$\forall\, a_1, a_2 \in A, \quad f(a_1) = f(a_2) \;\Rightarrow\; a_1 = a_2.$$

Equivalently, using the contrapositive:

$$a_1 \neq a_2 \;\Rightarrow\; f(a_1) \neq f(a_2).$$

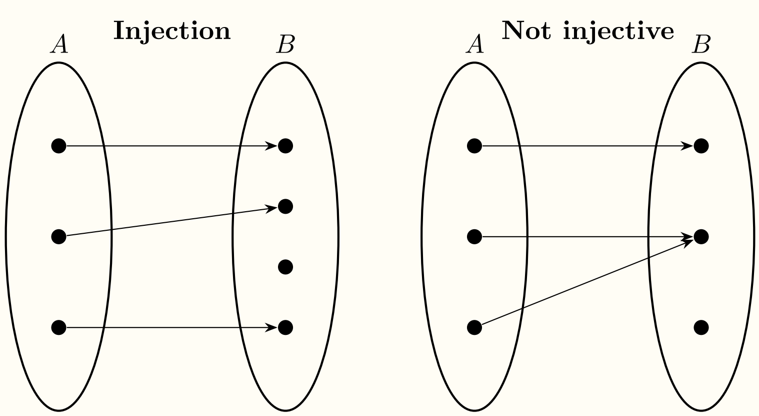
That is, distinct inputs always map to distinct outputs.

Remark. The two formulations are logically equivalent by contraposition:

$$\big(f(a_1) = f(a_2) \Rightarrow a_1 = a_2\big) \iff \big(a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)\big).$$

The contrapositive form is often more intuitive: no two different inputs share the same output.

Diagram: Injection vs. Non-injection



3 Surjective Functions (Surjections)

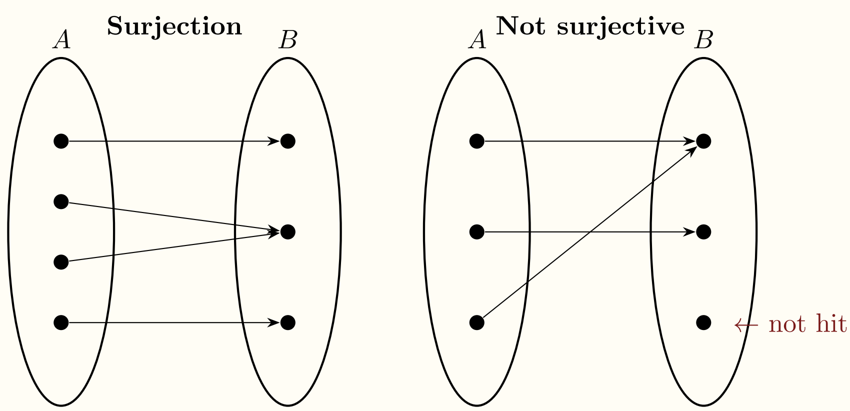
Definition 3.1 (Surjective function / surjection)**.** f is *surjective* (also called *onto* or a *surjection*) if and only if

$$\forall\, y \in B, \quad \exists\, x \in A, \quad f(x) = y.$$

That is, every element of the codomain B is the image of *at least one* element of A.

Remark. Surjectivity is a property of the *codomain*: it says that f "hits" every element of B. A function can be surjective without being injective (multiple inputs can share the same output, as long as no output is left uncovered).

Diagram: Surjection vs. Non-surjection



4 Comparing Injection and Surjection

Property	**Injective**	**Surjective**
Formal statement	$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$	$\forall y \in B,\ \exists x \in A,\ f(x) = y$
Informal meaning	No two inputs share an output	Every output is reachable
Constraint on	*inputs* (domain A)	*outputs* (codomain B)
Violations	Two arrows into one dot in B	A dot in B with no arrow

5 Bijections

Definition 5.1 (Bijection)**.** $f : A \to B$ is a *bijection* (or *one-to-one correspondence*) if it is both injective and surjective:

$$f \text{ is bijective} \iff f \text{ is injective} \wedge f \text{ is surjective.}$$

Remark. A bijection pairs every element of A with exactly one element of B and vice versa.

Bijections are precisely the functions that have a well-defined inverse $f^{-1} : B \to A$.

6 Cardinality

6.1 Comparing the sizes of sets

Given sets A and B, we compare their cardinalities using injections and surjections — this works even for infinite sets where direct counting is impossible.

Definition 6.1 ($|A| \leq |B|$)**.** The *cardinality of A is less than or equal to the cardinality of B*, written $|A| \leq |B|$, if and only if there exists an injective function from A to B:

$$|A| \leq |B| \iff \exists f\big(f : A \to B \ \wedge\ f \text{ is injective}\big).$$

Definition 6.2 ($|A| \geq |B|$)**.** The *cardinality of A is greater than or equal to the cardinality of B*, written $|A| \geq |B|$, if and only if there exists a surjective function from A to B:

$$|A| \geq |B| \iff \exists f\big(f : A \to B \ \wedge\ f \text{ is surjective}\big).$$

Remark. The asymmetry is intuitive: an injection $A \hookrightarrow B$ "fits" all of A into B without collisions, so B must be at least as large. A surjection $A \twoheadrightarrow B$ "covers" all of B from A, so A must be at least as large.

6.2 Equal cardinality

Definition 6.3 ($|A| = |B|$)**.** A and B have the *same cardinality*, written $|A| = |B|$, if and only if *all three* of the following equivalent conditions hold:

$$\begin{aligned} |A| = |B| &\iff \exists f\big(f : A \to B \ \wedge\ f \text{ is injective}\big) \ \wedge\ \exists g\big(g : A \to B \ \wedge\ g \text{ is surjective}\big) \\ &\iff \exists f\big(f : A \to B \ \wedge\ f \text{ is bijective}\big). \end{aligned}$$

That is, $|A| = |B|$ means both $|A| \leq |B|$ and $|A| \geq |B|$, which is equivalent to the existence of a bijection.

Theorem 6.1 (Cantor–Schröder–Bernstein)**.**

$$\forall x\, \forall y\, \Big(\big(|x| \leq |y| \ \wedge\ |y| \leq |x|\big) \ \Rightarrow\ |x| = |y|\Big).$$

That is, if there exist injections $f : x \to y$ and $g : y \to x$, then there exists a bijection between x and y.

Remark (This is one direction only)**.** The theorem is stated as $\Rightarrow$, not $\iff$. The converse direction —

$$|x| = |y| \;\Rightarrow\; |x| \leq |y| \;\wedge\; |y| \leq |x|$$

— is *trivially true by definition*: since $|x| = |y|$ means a bijection $f : x \to y$ exists, and a bijection is both injective and surjective, we immediately get injections in both directions. The non-trivial, hard direction is the one the theorem proves: that *two injections* (which need not be inverses of each other) are sufficient to guarantee a bijection exists.

6.3 Summary table

Notation	**Meaning**	**Witness required**
$\lvert A\rvert \leq \lvert B\rvert$	A is no larger than B	injective $f : A \to B$
$\lvert A\rvert \geq \lvert B\rvert$	A is no smaller than B	surjective $f : A \to B$
$\lvert A\rvert = \lvert B\rvert$	A and B are the same size	injection *and* surjection $A \to B$;
		equivalently, bijection $f : A \to B$

7 Summary

1. **Injection**: $\forall a_1, a_2 \in A,\; f(a_1) = f(a_2) \Rightarrow a_1 = a_2$.
2. **Surjection**: $\forall y \in B,\; \exists x \in A,\; f(x) = y$.
3. **Bijection**: injective $\wedge$ surjective.
4. $|A| \leq |B| \iff \exists$ injective $f : A \to B$.
5. $|A| \geq |B| \iff \exists$ surjective $f : A \to B$.
6. $|A| = |B| \iff \exists$ injective $f : A \to B \wedge \exists$ surjective $g : A \to B \iff \exists$ bijection $f : A \to B$.

Remark (Looking ahead)**.** We will use injections and surjections to compare the sizes of infinite sets (cardinality) and define countability.